

Cooperative Multi-agent Reinforcement Learning for Fair Dynamic Pricing

Vishwanath Vinod¹ and Rachel Kalpana Kalaimani²

Indian Institute of Technology Madras, Chennai 600036, India
vishwanathvinod2004@gmail.com¹
rachel@ee.iitm.ac.in²

Abstract. Dynamic pricing optimizes revenue by adjusting prices in real-time based on market demand and consumer behaviour. By capitalizing on real-time adjustments that account for differing willingness to pay across customer segments, dynamic pricing can lead to unequal pricing strategies that disproportionately benefit specific groups based on their demand characteristics. Fairness is introduced as an additional constraint in dynamic pricing algorithms to mitigate disparities and avoid discrimination against specific demographics. Integrating fairness into dynamic pricing promotes ethical practices and contributes to sustainable success. This paper presents a novel adaptation of existing Cooperative Multi-Agent Reinforcement Learning frameworks for fair dynamic pricing within a finite inventory and limited selling horizon setting. We define an innovative dual-reward structure that assigns the profit of each group as the individual reward for each agent, while fairness, quantified using Jain’s Index, is modelled as the team reward. We conduct experiments in a simulated environment and compare our algorithm with existing Reinforcement Learning algorithms that use a mixed reward signal for fairness and profit. The results demonstrate that this dual-reward structure converges to a better optimum, achieving higher levels of fairness and profit while facilitating faster convergence. We also introduce a control parameter that allows the model to effectively regulate the trade-off between fairness and profitability by adjusting the relative weight of the dual reward signals in the policy gradient step. This allows for flexible prioritization of either objective based on the desired outcome.

Keywords: Dynamic Pricing, Multi-Agent Reinforcement Learning, Fairness, Jain’s index

1 Introduction

Dynamic pricing is a pricing strategy where the price of a product or service fluctuates based on current market conditions [1]. This strategy has gained considerable traction across various industries due to its potential for revenue maximization by leveraging real-time data to adjust prices based on factors such

as customer willingness to pay, competitor pricing, fashion trends, seasonality, remaining inventory, and the time remaining in the selling horizon. With the advent of E-commerce platforms, dynamic pricing strategies have gained importance [2] and firms increasingly deploy algorithms like personalised pricing, surge pricing, price skimming, penetration pricing, customer-segmentation pricing and competitive pricing to observe customers and set prices dynamically [3]. These conventional dynamic pricing strategies often depend on rigid rule-based algorithms, which limit their adaptability to fluctuating market conditions and customer behaviour. Later, second-generation algorithms based on Deep Reinforcement Learning (DRL) gained prominence due to their adaptive control in response to varying market conditions, incorporating both immediate rewards, such as current profits, and long-term outcomes like customer loyalty and market share. This approach enables DRL algorithms to learn intricate, context-sensitive pricing strategies that flexibly respond to real-time market changes [4].

However, while these pricing strategies effectively optimize revenue, they can inadvertently lead to discriminatory practices by adjusting prices based on various customer segments’ perceived willingness to pay. The widespread availability of customer data has intensified the use of personalized dynamic pricing (PDP), where companies leverage past purchase behaviour, browsing history, and personal attributes to predict consumer valuations [5]. This approach risks unequal treatment of different groups due to historical disadvantages or unobserved factors, potentially exacerbating socio-demographic disparities. Extensive studies have shown that pricing platforms based on discriminatory pricing regimes are bound to fail due to the notion of ‘self-centred equity aversion’ [6] among customers, the tendency of individuals to be averse to situations where they perceive themselves to be at a disadvantage relative to others. Such unfair pricing policies impart profoundly negative perceptions to the general consumer base and may result in a downturn in popularity, sales, and consequently profits [7]. From a societal standpoint too, price discrimination based on demographic or economic markers is undesirable for a fair and egalitarian society. To address this, the notion of group fairness [8], which requires different groups of individuals to have statistically similar output distributions, must be incorporated into dynamic pricing. It acts as an additional constraint to ensure demographic parity and prevent discriminatory pricing.

In this paper, we propose a dynamic pricing approach that leverages a Cooperative Multi-Agent Reinforcement Learning (CoMARL) framework to achieve group fairness across diverse customer groups, each characterized by differing price-demand response curves, while simultaneously maximizing revenue. We also incorporate finite inventory and limited selling horizon constraints to ensure adaptability to limitations on resources and time. This framework has practical applications in many pricing situations including, but not limited to, e-commerce, airline ticket pricing, dynamic taxi fare calculation during rush hours, and event ticket pricing for concerts and sporting events.

Key Contributions

1. We address the problem of integrating fairness in dynamic pricing scenarios characterized by diverse customer distributions, incorporating additional constraints of finite inventory and a limited selling horizon. These additional constraints ensure that the pricing strategy is responsive to both resource availability and time constraints while simultaneously optimizing for fairness and profit. We propose to employ a Cooperative Multi-Agent Reinforcement Learning (CoMARL) framework to ensure demographic parity and prevent discriminatory pricing across dynamic pricing applications.
2. While prior literature combines fairness and profit into a single mixed reward signal, this study is, to our knowledge, the first to introduce a dual-reward system utilizing separate reward signals for profit and fairness. This enables a more effective optimization of the two competing objectives within the defined constraints (refer Section 4.4, Remark 1).
3. We perform a comprehensive empirical analysis of our CoMARL framework, benchmarking its performance against existing RL-based methods in dynamic pricing that utilize a mixed reward signal. Our experiments demonstrate that the proposed CoMARL framework not only surpasses these methods in terms of fairness and total profit but also achieves faster convergence (refer Section 6.1, Fig. 4).
4. We introduce a control parameter that governs the balance between fairness and profit by varying the relative weight of each reward signal in the policy gradient step. By adjusting this parameter, the model can determine the degree to which one objective is prioritized over the other, emphasising on fairness at the potential cost of profitability or vice versa (refer Section 6.2, Fig. 5).

2 Related Work

2.1 Dynamic Pricing

A comprehensive review of dynamic pricing strategies is presented in [9]. [10] further applies these strategies in finite inventory and limited selling horizons. [11] introduces an RL algorithm aimed at optimizing revenue, while [12] explores the use of Deep Q-Learning Networks (DQN) for perishable products within similar constraints. Aside from these, RL has been applied in various contexts to the broad problem of dynamic pricing [13]. Multi-agent RL (MARL) frameworks have recently been introduced in [14] to enhance cooperative pricing strategies for multiple perishable products.

2.2 Fairness

Fairness has been a focal point of research across multiple domains, including transportation systems [15], network design [16] and advertising [17], aiming

to ensure equitable access to resources and balanced outcomes. [18] provides a comprehensive survey on fairness, exploring specialized fairness metrics such as Max-Min fairness, Jain’s index and Gini index. Recent works in dynamic pricing model fairness as a utility trade-off with profit or a hard constraint preventing optimal profit-maximizing actions. For instance, [19] introduces four types of fairness: price, demand, surplus, and no-purchase valuation, highlighting the difficulty of achieving multiple fairness objectives simultaneously with deterministic pricing. [20] adopts a methodology of segmenting customers into distinct groups to address group fairness by minimizing the price gap between these segments, reflecting a similar approach employed in our work. Further, [21] introduces a mixed reward signal-based approach that integrates group fairness using Rotated Jain’s Index.

3 Background

Reinforcement Learning RL is an area of machine learning that studies how an agent takes actions in an environment to optimize a given reward signal [22]. RL differs from supervised learning as the agent learns by interacting with the environment, as opposed to learning with labelled data. The environment is often modelled as an MDP or Markov Decision Process defined by a set of states $\mathcal{S}$, actions $\mathcal{A}$, reward function $\mathcal{R}$, and transition probabilities $\mathcal{P}$. RL algorithms can be broadly classified into value-based methods and policy-based methods. In policy-based methods, such as policy gradient algorithms, the policy $\pi_\theta(a|s)$ is explicitly parameterized, and the parameters θ are optimized to maximize the expected cumulative reward, $J(\theta)$. Actor-critic algorithms improve upon this by incorporating a value function to reduce variance in policy updates.

Multi-Agent RL MARL [23] extends the RL framework to environments with multiple interacting agents, each governed by its own policy. These environments are often modelled as Partially Observable MDPs (POMDPs), where the joint state of all agents influences their decision-making processes. Cooperative MARL (CoMARL) [24] studies multiple agents collaborating to optimize a shared objective, using their collective actions and experiences to enhance overall system performance. In this paper, we build on Deep Deterministic Policy Gradients (DDPG) [25], an actor-critic algorithm that operates as an off-policy method. The DDPG algorithm extended to MARL settings results in the Multi-Agent DDPG (MADDPG), which allows each agent to learn individual policy by optimizing the gradient of the action-value function, $Q_i(s, a)$, which incorporates states $s = (o_1, \dots, o_N)$ and actions $a = (a_1, \dots, a_N)$ of all agents. The policy gradient for agent i at a given time step is:

$$\nabla_{\theta_i} J(\theta_i) = \mathbb{E}_{s, a \sim \mathcal{B}} [\nabla_{\theta_i} \pi_i(a_i | o_i) \nabla_{a_i} Q_i(s, a)] \quad (1)$$

where $\pi_i(\cdot)$ is the policy parametrized by θ_i , o_i is the local observation, and $Q_i(s, a_1, \dots, a_N)$ is the centralized action-value function for agent i . The experi-

ence replay buffer $\mathcal{B}$ is used to record experiences of all agents. We use a modified MADDPG framework adapting it to a CoMARL setting with a dual-reward mechanism $r = (r_1, \dots, r_N, \hat{r})$, where $r \in R : S \times A \times S \rightarrow \mathbb{R}^{N+1}$, such that each r_i is the individual reward for agent i , and $\hat{r}$ is the shared team reward. While each agent strives to maximize its individual reward, it also collaborates with other agents to enhance the overall team reward, thereby fostering a coordinated decision-making process that balances individual and collective objectives.

4 Problem Formulation

We focus on the problem of integrating group fairness into a dynamic pricing framework characterized by finite inventory and a limited selling horizon, while simultaneously optimizing total profit. To effectively capture the diverse demand behaviours of customers, we divide the customers into distinct groups, each characterized by a unique demand function. We denote these groups as $G = \{1, 2, 3, \dots\}$, and $|G|$ denotes the number of groups. Our setup imposes a finite inventory constraint, with items shared evenly across customer groups. This equitable distribution ensures equal access for customers of all groups, eliminates inventory differences as a source of unfairness, and leaves price as the sole factor driving disparities [19]. The limited selling horizon, $\mathcal{H}$, is implemented by capping the total number of days in the entire selling season. To ensure realistic pricing, the proposed prices, p_g for each group $g \in G$ are restricted to a suitable range $[p_l, p_h]$. The individual aspects of the problem statement are covered in the subsequent sections.

4.1 Demand

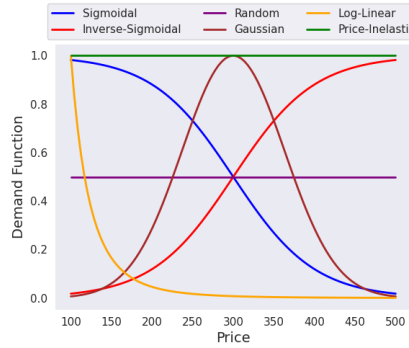


Fig. 1: Demand functions of Groups: In our experiments, we set the price range to $[100, 500]$ and select the demand function parameters such that these functions remain within $[0, 1]$ for this price range.

We adopt the demand models proposed in [10] and [21], where each group's demand at a given price (p_g) is modelled as a Bernoulli random variable ($D_g(p_g)$),

taking a value of 1 if a customer accepts this price and 0 if they reject it. The probability of acceptance is given by the value of group-specific demand function at that price ($\mathbb{D}_g(p_g)$). This probabilistic approach given by (2) is used to model the stochastic nature of customer behaviour.

$$D_g(p_g) \sim \text{Bernoulli}(\mathbb{D}_g(p_g)) \quad \forall p_g \in [p_l, p_h], g \in G \quad (2)$$

By assigning unique demand functions for each customer group, we effectively capture the diverse purchasing behaviours of customers. Table 1 presents an overview of the demand functions implemented in our experiments, outlining their mathematical formulations and associated parameter values. Fig. 1 illustrates these demand functions.

Table 1: Demand Functions for Customer Groups

Group	Customer Behaviour	Demand Function ($\mathbb{D}_g(p)$)
Group 1	Sigmoidal Demand	$\left(1 + e^{-(b+w \cdot p)}\right)^{-1}$
Group 2	Inverse Sigmoidal Demand	$\left(1 + e^{(b+w \cdot p)}\right)^{-1}$
Group 3	Log-Linear Demand	$e^{-(b' + w' \cdot \ln(p))}$
Group 4	Price Inelastic Demand	1 (constant)
Group 5	Gaussian Demand	$e^{-\left(\frac{(p-\mu)^2}{2\sigma^2}\right)}$
Group 6	Random Demand	0.5 (constant)

$$\begin{array}{lll} \textbf{Parameters:} & b = 6 & w = -0.02 \quad \mu = 300 \\ & b' = -20.44 & w' = 4.44 \quad \sigma^2 = 8000 \end{array}$$

Group 1 comprises customers who follow the standard inverse relationship, where demand increases as prices decrease. Group 2 has customers who associate higher prices with better quality, resulting in greater demand for higher-priced products. Group 3 represents more frugal customers who will only accept the product if offered at a low price. Group 4 includes customers who are indifferent to price and readily accept any offered amount, which leads to stable demand irrespective of the product's cost. Group 5 comprises customers who gravitate toward prices concentrated around the mean, reflecting a tendency to accept intermediate quality at a reasonable cost, balancing value and affordability. Group 6 consists of whimsical customers who make random purchasing decisions, akin to a coin toss, without a predetermined strategy, leading to demand independent of price.

4.2 Fairness Index

In this paper, we adopt Jain's Index (f_{jain}), proposed in [26] and widely used in resource allocation theory and dynamic pricing as a robust mathematical basis

for fairness. This index is adaptable to any number of customer groups, making it well-suited to our setting. It ranges from 0 to 1, with higher values indicating greater fairness.

The Jain index is given by:

$$f_{jain}(p_1, p_2, \dots, p_{|G|}) = \frac{\left(\sum_{g \in G} \bar{p}_g\right)^2}{|G| \sum_{g \in G} \bar{p}_g^2} \quad (3)$$

where $\bar{p}_g$ represents the average price allocated in group g . We prioritize ensuring fairness at each time step of the environment rather than calculating it over the entire episode. This is crucial because fairness calculated only at the episode level spans many interactions and may obscure individual unfair transactions. Even a few unfair interactions could contribute to a negative perception of pricing among customers, and the episodic average might not reflect these discrepancies. By focusing on fairness at each step, we ensure that the pricing remains equitable throughout the selling period, thereby enhancing customer satisfaction.

4.3 Customer Interaction

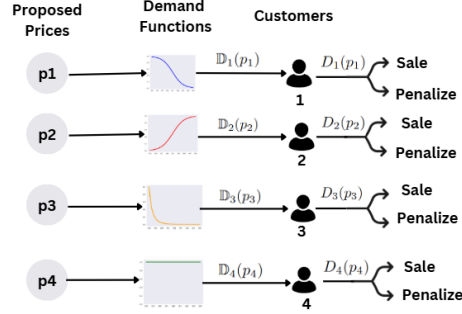


Fig. 2: Customer Interaction modelled for 4 Groups

At each time step t in $\mathcal{H}$, one customer from each group interacts independently with the pricing strategy. The customer influx rate, $\mathcal{C}$, representing the number of customers interacting per day, is held constant throughout the selling season to maintain focus on the demand functions of each group as the primary variable. While the constant customer influx ensures transactions occur at a uniform rate, the number of successful sales varies over time, depending on the group-specific demand functions and the prices offered at each timestep. If the price is accepted, a sale is recorded for that transaction. Otherwise, a penalty is imposed on the algorithm to discourage proposing prices that are unlikely to lead to a sale,

leaving unsold inventory. The dynamics of customer interaction is formalized in (4) and depicted in Fig. 2.

$$Interaction : \begin{cases} Sale & \text{if } D_g(p_g) = 1 \\ Penalty & \text{if } D_g(p_g) = 0 \end{cases} \quad (4)$$

4.4 Environment

A custom MARL environment is developed to model a dynamic pricing framework for multiple customer groups. We segment the customer base into distinct groups, each with a unique demand function, and train a dedicated RL agent for each group to learn an optimal pricing strategy. This multi-agent approach allows independent adaptation to the specific characteristics of each group. While multiple agents may pose challenges to group fairness, our method emphasizes group fairness by incorporating fairness measures into the reward structure. At each step, every agent simultaneously interacts with a customer from its respective group, proposing a price based on its learned strategy. The episode, representing an entire selling season with a limited horizon, concludes after the designated number of days, or when the inventory for all agents is fully exhausted.

The environment parameters are:

State: The environment’s state space comprises the current inventory level and days left in the selling season of all agents, $s = \{(d_1, n_1), \dots, (d_{|G|}, n_{|G|})\}$, where d_i denotes the number of days remaining in the selling season, and n_i represents the remaining inventory at any given moment.

Action: The environment’s action space corresponds to the proposed price at each time step for customers in each group normalized to the range $[0,1]$. These are given as $a = \{\tilde{p}_1, \dots, \tilde{p}_{|G|}\}$, where $\tilde{p}_g$ is the normalized proposed price for customers from group g .

Reward: Existing RL algorithms (refer Appendix) for fair dynamic pricing tackle the challenge of balancing fairness and profit by using a mixed reward signal for each agent, which integrates both fairness and profit rewards into a single metric. The hyperparameters β_p and β_f control the weight assigned to the two objectives, i.e., profit and fairness.

This mixed reward signal is modelled as:

$$r_g = \begin{cases} \beta_p \cdot \tilde{p}_g + \beta_f \cdot f_{\text{fairness}}(p_1, \dots, p_{|G|}) & \text{if Sale} \\ -0.5 & \text{if Penalty} \end{cases} \quad (5)$$

Dual-Reward structure: Our approach involves decomposing the reward into two distinct signals: an individual reward for each agent, representing its profit, and a collective team reward that promotes fairness, derived from the joint actions (i.e., proposed prices of all agents).

The individual reward for each agent is modelled as:

$$r_g = \begin{cases} \tilde{p}_g, & \text{if Sale} \\ -0.5, & \text{if Penalty} \end{cases} \quad (6)$$

The team reward for all agents is modelled as:

$$\hat{r} = f_{\text{jain}}(p_1, \dots, p_{|G|}) \quad (7)$$

This dual-reward structure optimizes both reward functions more efficiently, providing better control over the trade-off between the competing objectives of fairness and profit as explained in the following remark.

Remark 1. [27] demonstrates through the defensive-escort problem that entangled reward functions become unfactorizable during training, causing the learning to oscillate between optimizing either of the objectives, resulting in sub-optimal and unstable solutions. Existing RL algorithms conflate fairness, a collective team reward, with profit, an individual agent’s reward and fail to adequately distinguish the distinct impacts of their actions on each objective. Our Dual reward algorithm provides a clear separation of profit and fairness reward signals, preventing agents from inadvertently prioritizing one objective over the other and converging to sub-optimal outcomes.

5 Algorithm

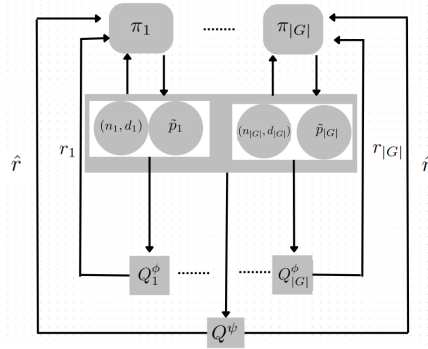


Fig. 3: Dual-Critic Architecture: A global centralized critic (Q^ψ), shared by all agents, estimates fairness (global reward) based on the entire state and action space. Local critics (Q_g^ϕ), specific to each agent estimate the profit (local reward) using only that agent’s state and action.

We propose a pricing strategy based on a modified Decomposed MADDPG (Dec-MADDPG) algorithm [27], specifically designed to balance the dual objectives of maximizing individual profits and group fairness. A key feature of this method is the dual-critic architecture, illustrated in Fig. 3, which separates the learning processes for global and local rewards. These rewards are integrated in the policy gradient step of our algorithm¹ where we introduce an additional parameter, λ , to control the trade-off between fairness and profitability by adjusting the relative weight of each reward signal instead of using a weighted reward function.

Algorithm 1: Dual-Reward Algorithm (DRA)

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1: for episode = 1 to  $\mathcal{E}$  do ▷  $\mathcal{E}$ : Total Episodes
2:   for  $t = 1$  to  $\mathcal{H}$  do ▷  $t$ : Time step
3:     Get state  $s^t = ((d_1^t, n_1^t), \dots, (d_{|G|}^t, n_{|G|}^t))$ 
4:     for groups  $g = 1$  to  $|G|$  do
5:       Sample noise  $\eta^t \sim \text{OU}(\theta, \mu, \sigma)$  given by (9)
6:       Propose price  $p_g^t = \pi_g^{\theta_g}((d_g^t, n_g^t)) + \eta$ 
7:       Clip and normalize to get action  $(\tilde{p}_g^t)$ 
8:       Compute global reward  $\hat{r}^t$  using (7)
9:       for groups  $g = 1$  to  $|G|$  do
10:        if  $n_g^t > 0$  then ▷ Inventory check
11:          Compute local rewards  $r_g^t$  using (6)
12:          if it is a sale then
13:             $n_g^t = n_g^t - 1$  ▷ Update inventory
14:          if  $t \bmod \mathcal{C} = 0$  then
15:             $d_g^t = d_g^t - 1$  ▷ Update days left
16:          Store updated state  $s^{t'} = (n_g^t, d_g^t)$  as next state
17:          Store  $(s^t, a^t, \hat{r}^t, r^t, s^{t'})$  in  $\mathcal{B}$  ▷  $\mathcal{B}$ : Buffer
18:        if episode  $> M$  then
19:          GlobalCriticUpdate()
20:          LocalCriticUpdate()
21:          Update policy parameters  $(\theta_g)$  using (8)
22:        if  $t \bmod t_0 = 0$  then
23:           $\theta_g^{tg} \leftarrow \tau \theta_g + (1 - \tau) \theta_g^{tg}$  ▷ Soft Update

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The policy gradient of each agent in conventional MARL like the MADDPG algorithm is calculated according to (1). Our approach shifts each agent's policy $(\pi_g^{\theta_g})$ in a direction that maximizes both the global and local rewards.

¹ Exploration is facilitated through an initial exploratory phase of M episodes and by adding noise (9) in action selection. Learning stability is enhanced by using target networks for policy (θ_g^{tg}) , local critics (ϕ_g^{tg}) and global critic (ψ^{tg}) , which are soft updated (τ) every t_0 steps.

Subroutine 1: Global Critic Update

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- 1: Sample mini-batch of size S : $(s^j, a^j, \hat{r}^j, s^{j'})$ from $\mathcal{B}$
 - 2: Compute next action $a^{j'}$ using target network $(Q^{\psi^{tg}})$
 - 3: Set $Y^j = \hat{r}^j + \gamma \cdot Q^{\psi^{tg}}(s^{j'}, a^{j'})$ ▷ Q-learning
 - 4: Update global critic parameters (ψ) by minimizing

$$\frac{1}{S} \sum_j \left(Y^j - Q^\psi(s^j, a^j) \right)^2$$

- 5: **if** $t \bmod t_0 = 0$ **then**
 - 6: $\psi^{tg} \leftarrow \tau \psi + (1 - \tau) \psi^{tg}$ ▷ Soft Update
-

The policy gradient for each agent is formulated in (8).

$$\begin{aligned} \nabla J(\theta_g) = & \lambda \cdot \overbrace{\mathbb{E}_{s,a \sim \mathcal{B}} [\nabla_{\theta_g} \pi_g(\tilde{p}_g | (d_g, n_g)) \nabla_{\tilde{p}_g} Q^\psi(s, a)]}^{\text{Global Critic}} \\ & + \underbrace{\mathbb{E}_{(d_g, n_g), \tilde{p}_g \sim \mathcal{B}} [\nabla_{\theta_g} \pi_g(\tilde{p}_g | (d_g, n_g)) \nabla_{\tilde{p}_g} Q_g^{\phi_g}((d_g, n_g), \tilde{p}_g)]}_{\text{Local Critic}} \end{aligned} \quad (8)$$

Subroutine 2: Local Critic Update

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- 1: **for** groups $g = 1$ to $|G|$ **do**
 - 2: Sample mini-batch of size S : $(s^j, a^j, r^j, s^{j'})$ from $\mathcal{B}$
 - 3: Compute next action $\tilde{p}_g^{j'}$ using target network $(Q_g^{\phi_g^{tg}})$.
 - 4: Set $y_g^j = r_g^j + \gamma \cdot Q_g^{\phi_g^{tg}}((d_g^{j'}, n_g^{j'}), \tilde{p}_g^{j'})$ ▷ Q-learning
 - 5: Update local critic parameters (ϕ_g) by minimizing

$$\frac{1}{S} \sum_j \left(y_g^j - Q_g^{\phi_g}((d_g^{j'}, n_g^{j'}), \tilde{p}_g^{j'}) \right)^2$$

- 6: **if** $t \bmod t_0 = 0$ **then**
 - 7: $\phi_g^{tg} \leftarrow \tau \phi_g + (1 - \tau) \phi_g^{tg}$ ▷ Soft Update
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The implementation of this algorithm, network architecture and the parameters used are provided in the Appendix.

6 Experimental Results

The main motivation behind our experiments was to evaluate the performance of our dual-reward algorithm relative to conventional RL methods that utilize mixed reward signals. Evaluations were conducted in a MARL environment simulating dynamic pricing scenarios, characterized by varied customer distributions

(as detailed in Table 1) along with constraints on inventory and time. To ensure the robustness and scalability of our algorithm, we conducted experiments across a selection of market scenarios with varied customer base structures. We analyze the net profit, group fairness (quantified by Jain’s index) and speed of convergence of the two algorithms. We also evaluate the effectiveness of the control parameter (λ) in balancing the competing objectives of fairness and profit through experiments across various settings of λ . Key environmental parameters like the number of days (30), total inventory ($300 \times |G|$), and number of customers ($600 \times |G|$), were kept constant for these experiments.

6.1 Comparative Performance Analysis

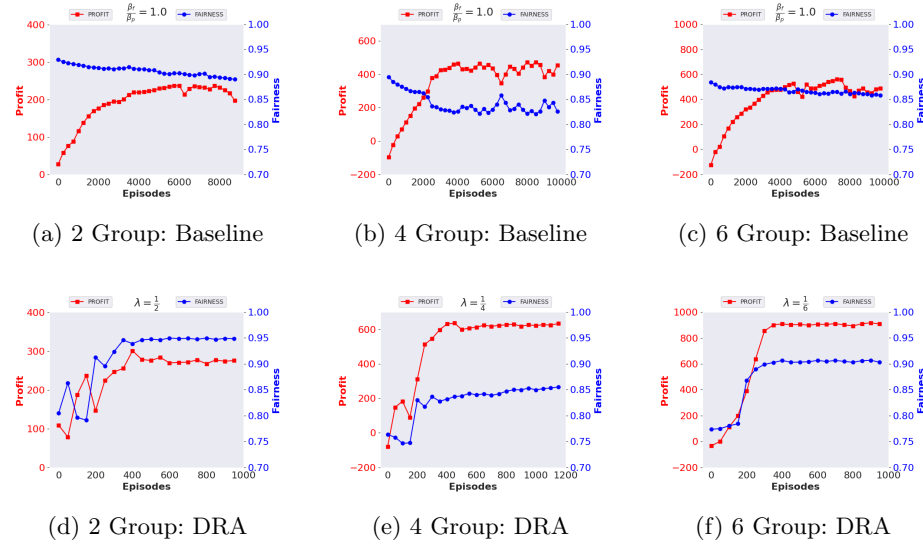


Fig. 4: Comparison of profit and fairness against baseline for selected customer distributions consisting of 2 (Group 1 & 2), 4 (Group 1,2,3 & 4), and 6 (Group 1,2,3,4,5 & 6) groups (Refer Table 1) with $\frac{\beta_f}{\beta_p} = 1.0$ and $\lambda = \frac{1}{|G|}$. The Dual Reward Algorithm (DRA) achieves higher levels of profit and fairness while facilitating faster convergence.

Fig. 4 illustrate the performance comparison of our dual-reward algorithm against a baseline that utilizes a mixed reward signal across customer bases consisting of 2 (Group 1 & 2), 4 (Group 1,2,3 & 4), and 6 (Group 1,2,3,4,5 & 6) groups, respectively, following the demand functions outlined in Table 1. We conducted simulations across multiple values of λ to evaluate the performance of our algorithm against the baseline algorithm (refer Appendix), using the benchmark setting of $\frac{\beta_f}{\beta_p} = 1.0$ from [21]. We observe that $\lambda = \frac{1}{|G|}$ consistently outperforms

the benchmark in both profit and fairness across all three scenarios, making it a good choice for comparison. This choice of λ is also intuitive since it ensures that the global critic’s influence scales with the number of agents ($|G|$), preventing it from overpowering the individual learning dynamics of the agent while still reinforcing cooperation.

Speed of Convergence

These results also highlight the accelerated convergence of our dual-reward algorithm. In all tested scenarios, the dual-reward algorithm achieves convergence within 500 episodes, whereas the mixed-reward approach requires over 4000 episodes. Faster convergence is crucial in dynamic pricing settings for higher cumulative rewards and high responsiveness to changing market conditions.

6.2 Effect of Control parameter (λ)

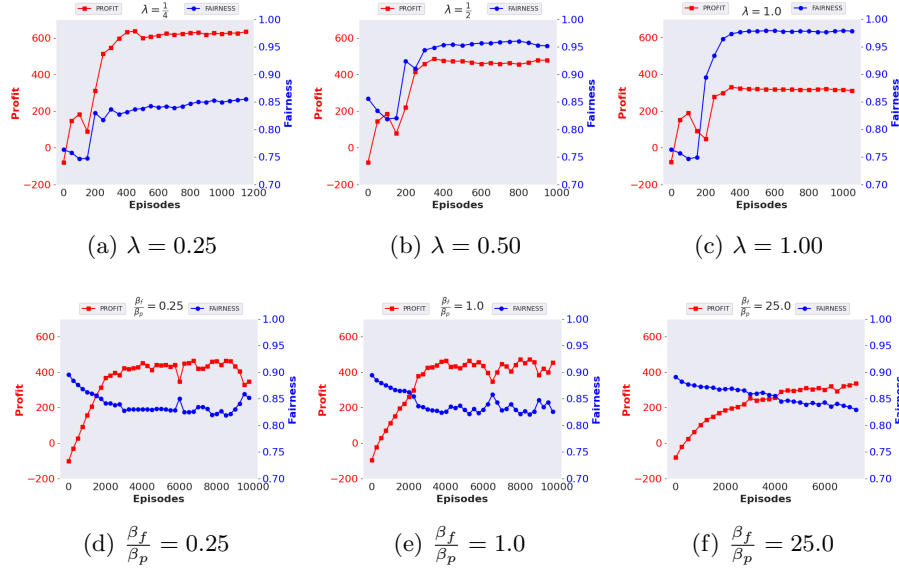


Fig. 5: Comparing effect of varying λ and weights β_f, β_p on the trade-off between fairness and profit in a 4 (Group 1,2,3 & 4) group setting. Increasing λ results in increased fairness at the cost of profit. Varying $\frac{\beta_f}{\beta_p}$ shows no distinctive trend.

In traditional multi-objective optimization, weights are assigned to rewards based on the priority of each objective as shown in (5). We improve upon this by introducing a control parameter (λ) into the framework of our algorithm during the policy gradient step. This enables better control over the trade-off between the competing objectives of maximizing fairness and profit. Table 2 consolidates the mean and standard deviation of outcomes over three experiments for each

setting of the control parameter (λ). These results highlight the gradual shift in priority between the fairness and profit objectives as λ is varied for different customer base settings. Notably, increasing the value of λ results in greater emphasis on fairness at the cost of profit for all the selected customer bases. However, no such distinctive trend can be observed on varying the weights (β_p and β_f). In particular, increasing the relative weight of the fairness component in the reward doesn't correspond to an increase in emphasis on the fairness objective. Fig. 5 illustrates this comparison in the four customer group settings.

Table 2: Profit and Fairness after 1000 Episodes Averaged over Three Trials for Different Group and λ Settings

Groups (G)	λ	Profit	Fairness
1,2	$\frac{1}{ G }$	259.6 ± 20.1	0.952 ± 0.004
	1	105.7 ± 30.3	0.983 ± 0.006
1,2,3,4	$\frac{1}{ G }$	590.6 ± 49.9	0.819 ± 0.007
	0.5	391.1 ± 65.1	0.958 ± 0.005
	1	313.9 ± 8.3	0.981 ± 0.004
1,2,3,4,5,6	$\frac{1}{ G }$	917.2 ± 33.0	0.921 ± 0.008
	0.5	826.9 ± 30.1	0.934 ± 0.004
	1	622.2 ± 4.1	0.964 ± 0.001

7 Conclusion and Further Work

In this paper, we explore the problem of group fairness in dynamic pricing with finite inventory and limited selling horizon constraints. We adapt existing MARL algorithms, like Decomposed Multi-Agent DDPG, which allow optimization towards the dual objectives of fairness and profit maximization. Our empirical work shows that this adaptation is much more effective at optimizing both objectives, as well as in achieving very fast convergence to a well-performing and fair pricing policy. Our novel use of a control parameter which weights individual and team rewards in MARL, improves the robustness of our method making it adaptable to a number of scenarios requiring different levels of strictness in terms of the fairness constraint.

There exist several potential avenues for future research on this problem. The assumption of constant customer arrival can be relaxed to study a more realistic setting where the customer arrival is a deterministic function of the proposed price. Exploring distillation techniques in this MARL framework to transfer knowledge from the team policy to each agent's policy by some form of

divergence regularization or distributed optimization may also be beneficial. We look forward to exploring these and other interesting variations in future works.

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Appendix Experimental Setup

A.1 Baseline Algorithm

[21] models the reward function in similar settings as a Gaussian centered at the target profit and fairness, rewarding proximity to these target values. However, we employ a mixed reward signal that combines fairness and profit linearly, as shown in (5). The rest of the algorithm is adapted to our problem settings without notable modifications. We make this modification because we aim to achieve maximum fairness and profit. If we set a target value for fairness and profit as we don’t know the maximum possible profit achievable at a given value of fairness or vice versa we will essentially be constraining one of these objectives.

Baseline: Mixed Reward Signal

```

1: for  $episode = 1$  to  $\mathcal{E}$  do ▷  $\mathcal{E}$ : Total Episodes
2:   for  $t = 1$  to  $\mathcal{H}$  do
3:     for  $g = 1$  to  $|G|$  do ▷  $\epsilon$ -greedy action
4:        $p_g^t = \begin{cases} \arg \max_{p_g^t} Q_g^{\theta_g}((n_g^t, d_g^t), p_g^t), & \text{w.p. } 1 - \epsilon \\ \text{Random price,} & \text{w.p. } \epsilon \end{cases}$ 
5:       Clip and normalize to get action  $(\tilde{p}_g^t)$ 
6:       Calculate fairness using (3)
7:       Compute rewards for all agents  $\{r_g^t\}_{g=1}^{|G|}$  by (5)
8:       Store  $(s^t, a^t, r^t, s^{t'})$  in  $\mathcal{B}$  ▷  $\mathcal{B}$ : Buffer
9:       Sample mini-batch of size  $S$  from  $\mathcal{B}$ 
10:      for  $g = 1$  to  $|G|$  do
11:         $y_g^j = r_g^j + \gamma \arg \max_{a_g^j} Q_g^{\theta_g^g}(s_g^{j'}, \tilde{p}_g^{j'})$ 
12:        Update network parameters  $(\theta_g)$  by minimizing:
          
$$\frac{1}{S} \sum_{j=1}^S \left( y_g^j - Q_g^{\theta_g}(s_g^j, \tilde{p}_1^j, \dots, \tilde{p}_{|G|}^j) \right)^2$$

13:      if  $t \bmod t_0 = 0$  then ▷ Target Network
14:         $\theta_g^{tg} \leftarrow \theta_g$ 

```

A.2 Network Architecture:

The proposed algorithm employs three neural networks: the Actor, the Local Critic and the Global Critic². Each agent is equipped with its Actor and Local Critic network, while a single Global Critic is shared across all agents to facilitate coordinated learning. All networks are initialized using Xavier Uniform initialization.

Actor Networks Each Actor network takes as input the state of its corresponding agent and passes it through 2 hidden layers with 300 and 600 units, both employing the ReLU activation function. The output layer utilizes a tanh activation function to ensure that the generated actions are bounded within the predefined price range.

Local Critic Networks Each Local Critic network takes as input the state of its corresponding agent and the action taken by it. The state of the agent is passed as input in the first layer and the action is concatenated with the output of the first layer and passed as the input of the second layer. The network has 3 hidden layers of dimensions 512, 256 and 128. All the hidden layers as well as the output layer employ the ReLU activation function.

² **Network Parameters:** Replay Buffer Size = 10^6 , Batch Size = 256, $\gamma = 0.95$, $\tau = 0.01$, $M = 200$, $t_0 = 100$, Learning Rate = 10^{-4}

Global Critic Network The Global Critic network takes as input the states and actions of all the agents. The states of all the agents is passed as input in the first layer and the actions taken by all the agents is concatenated with the output of the first layer and passed as the input of the second layer. The network has 2 hidden layers of dimensions 1024 and 512 with ReLU activation function. The output layer employs the Linear activation function.

A.3 Noise (η^t)

During training, we encourage exploration by noising the deterministic actions produced by the policy network using a variant of Ornstein-Uhlenbeck (OU) noise, which generates temporally correlated noise and is well-suited for continuous action spaces. The following equation governs the evolution of this noise³:

$$\Delta\eta^t = \theta \cdot (\mu - \eta^t)\Delta t + \sigma\sqrt{\Delta t} \cdot \mathcal{N}(0, 1) \quad (9)$$

where θ represents the rate at which the noise reverts to the mean μ , σ controls the magnitude of the random fluctuations and is gradually reduced or annealed as training progresses until it reaches a specified minimum value σ_{min} , Δt is the time increment, and $\mathcal{N}(0, 1)$ denotes normal distribution with mean 0 and variance 1.

³ **Noise Parameters:** Annealing steps= 2×10^6 , $\Delta t = 0.01$, $\mu = 0.0$, $\theta = 0.15$, $\sigma = 160.0$, $\sigma_{min} = 5.0$